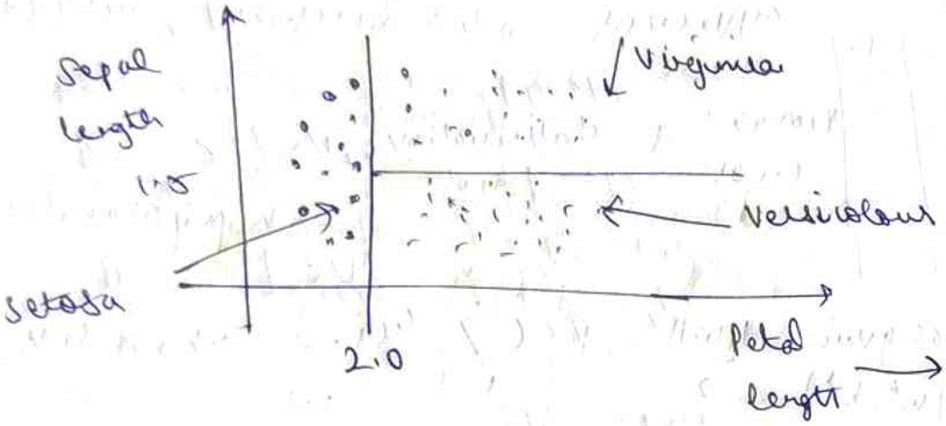
Decision Tree:

- Begin with training set having classification/regression
- Determine best feature to split the data on.
- Split data into subsets. This defines a node on a tree. node is a splitting point based on certain feature from data
- Recursively generate new nodes

Decision tree use hyperplanes $\parallel$ to one axis to cut coord system into hypercuboids.



Entropy $\longrightarrow$ measure of disorder.

$$E(S) = \sum_{i=1}^{c} -P_i \log_2 P_i$$

$P_i \rightarrow$ probabil of a class 'i' in data.

if only YES/NO,

$$E(S) = -P_Y \log_2(P_Y) - P_N \log_2(P_N)$$

Q)

Sal	Age	Purchase
20K	21	Y
10K	45	N
60K	27	Y
15K	31	N
12K	18	N

$E(S) = ?$

$P_Y = 2/5$, $P_N = 3/5$

$-2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.97$

- for a 2 class problem,

$E(S) \in [0, 1]$

if all Y/N, $-5/5 \log_2(5/5) + 0 = 0$

$n(Y) = n(N)$

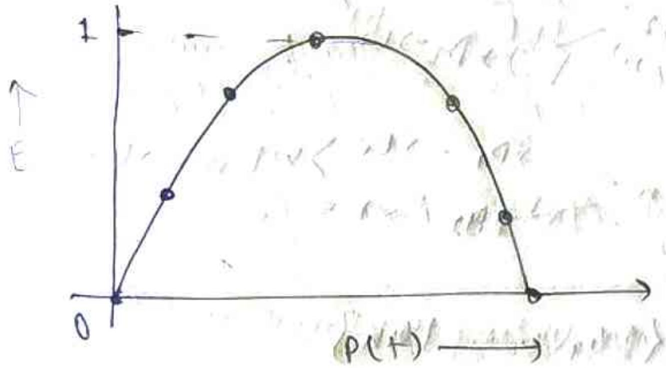
$-1/2 \log_2(1/2) - 1/2 \log_2(1/2) = -1/2(-1) - 1/2(-1)$

$= 1/2 + 1/2 = 1$

- for 3 class

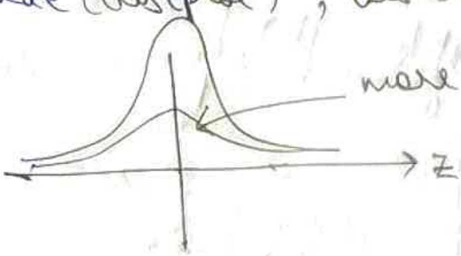
$E(S) \geq 0$ (No upper limit)

Entropy vs Probability:



for 2 continuous variables:

Plot kde (distplot), less the peak & more entropy.



Information gain: metric used to train decision trees.
While constructing a decision tree, find attribute with highest info gain.

$$IG = E(\text{parent}) - \underbrace{W_{avg} \cdot E(\text{children})}_{\text{weighted avg (weighted entropy)}}$$

Gini impurity:

$$[E = -P_Y \log_2(P_Y) - P_N \log_2(P_N)]$$

$$GI = 1 - (P_Y^2 + P_N^2)$$

Handling numerical data:

$D \longrightarrow f > v_1 \rightarrow D_1 \rightarrow E_1,\ D_2 \rightarrow E_2 \ \rangle\ WE_1 \rightarrow IG_1$

$D \longrightarrow f > v_2 \rightarrow D_1 \rightarrow E_1,\ D_2 \rightarrow E_2 \ \rangle\ WE_2 \rightarrow IG_2$

$\vdots$

$D \longrightarrow f > v_n \rightarrow D_1 \rightarrow E_1,\ D_2 \rightarrow E_2 \ \rangle\ WE_n \rightarrow IG_n$

v	D
v_1	Y
v_2	N
$\vdots$	Y
	N
	N

$$\max\{IG_1, \ldots, IG_n\}$$